

Axioms of Quantum Mechanics and the Arithmetic Structure of the Bost–Connes System

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Abstract

We analyze, from the arithmetic structure of the Bost–Connes (BC) system, the correspondence of each of the five von Neumann axioms of quantum mechanics within the BC framework. The main results are as follows: (I) C^* -positivity $\varphi(a^*a) \geq 0$ naturally gives rise to the Hilbert space $\ell^2(\mathbb{N}^+)$; (II) arithmetic function operators commuting with H describe self-adjoint observables; (III) the overlap relation between KMS states encodes the Born rule $P = |c_n|^2$; (IV) the strict monotonicity of $\zeta(\beta)$ determines the unitary time evolution $U(t) = e^{-itH}$; (V) the fundamental theorem of arithmetic $n = \prod p^{a_p}$ corresponds to the tensor product decomposition $|n\rangle = \bigotimes |a_p\rangle$.

1 Introduction

The axiomatic framework of quantum mechanics due to von Neumann [2] comprises five axioms (Dirac [10] provides a complementary perspective): (I) the state space is a Hilbert space; (II) observables are self-adjoint operators; (III) measurement probabilities are given by the Born rule; (IV) time evolution is a unitary group; (V) the state space of a composite system is a tensor product. In the standard formulation, these five axioms are regarded as mutually independent physical postulates.

In the present paper, we analyze these five axioms from the perspective of the Bost–Connes (BC) system [1]. The core content of all five axioms finds natural correspondences within the BC system. We do not invoke measurement, observers, or wavefunction collapse. The key insight is that the BC system, for $\beta > 1$, possesses a family of Galois-transitive KMS states $\{\varphi_{\beta,\alpha}\}_{\alpha \in \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})}$. Different KMS states agree on diagonal quantities (energy eigenvalues $n^{-\beta}/\zeta(\beta)$) but differ on off-diagonal quantities (phases). The Born rule emerges as the *unique overlap functional* between two such states: probability is not a property of a particle being “measured,” but a *relational property* between two KMS states that cannot fully observe each other’s phases.

The derivation proceeds in five steps:

1. The positivity of C^* -algebraic states naturally gives rise to an inner product ($p = 2$);
2. the partition function $Z(\beta) = \zeta(\beta)$ determines the normalization $w_n = 1$;
3. the Galois phase mismatch between KMS states yields the Born rule $P = |c_n|^2$;
4. the strict monotonicity of ζ determines a unique unitary group;
5. the fundamental theorem of arithmetic gives the tensor product.

2 The Bost–Connes System

The BC system is defined on the set of positive integers $\mathbb{N}^+$. Completion (from the rationals to the reals and then to the complexes) is a standard mathematical step required for the GNS construction [15].

Definition 1 (BC System). *The BC system is defined by the following data [1], divided into arithmetic input and derived structure:*

Arithmetic input (three items):

- (i) *The set of objects $\mathbb{N}^+ = \{1, 2, 3, \dots\}$.*
- (ii) *Unique factorization: $n = \prod_p p^{a_p(n)}$ (the fundamental theorem of arithmetic).*
- (iii) *The Hamiltonian $H(n) = \log n$ (the unique non-negative completely additive function on $\mathbb{N}^+$; Proposition 3).*

Derived structure (four items, each uniquely determined by (i)–(iii)):

- (iv) *The multiplication operator $\mu_m(n) = mn$ (faithful representation of the multiplicative action of $\mathbb{N}^+$; uniquely determined by (ii)).*
- (v) *The one-parameter automorphism group $\sigma_t(\mu_m) = m^{it}\mu_m$, $t \in \mathbb{R}$ (pointwise multiplication n^{it} on the function space over $\mathbb{N}^+$; uniquely determined by (iii); see Remark 2).*
- (vi) *The partition function $Z(\beta) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta)$, $\beta > 1$ [14] ($= \sum_n n^{-\beta}$ by direct summation; uniquely determined by (iii)).*
- (vii) *KMS states ($\beta > 1$): $\varphi_\beta(|n\rangle\langle m|) = \delta_{nm} n^{-\beta} / \zeta(\beta)$ (uniquely determined by the KMS condition for the dynamics σ_t ; this is the main theorem of [1]).*

Remark 2 (Uniqueness of the derived structure). *Items (iv)–(vii) are not independent assumptions; each is the unique mathematical consequence of (i)–(iii):*

- (iv): *The multiplicative action $m \cdot n = mn$ is the unique faithful action of $\mathbb{N}^+$ on itself compatible with unique factorization.*
- (v): *The map $\sigma_t(f)(n) = n^{it}f(n)$ is pointwise multiplication on the function space over $\mathbb{N}^+$ —a purely algebraic operation requiring no Hilbert space. Its uniqueness as a one-parameter automorphism group is established in Theorem 9 (Section 6).*
- (vi): *$Z(\beta) = \sum_n e^{-\beta \log n} = \sum_n n^{-\beta}$ is a direct sum from H ; no trace or inner product is needed.*
- (vii): *The KMS state at inverse temperature β is uniquely determined by the dynamics σ_t (unique for $\beta < 1$; a Galois-transitive family for $\beta > 1$ [1]; see [18] for number field generalizations).*

The BC algebra may be defined abstractly via generators μ_n , $e(r)$ ($r \in \mathbb{Q}/\mathbb{Z}$) and relations [1]; the concrete representation on $\ell^2(\mathbb{N}^+)$ is the output of the GNS construction (Theorem 4), not the input. The C^ -algebraic language provides the mathematical framework.*

Proposition 3 (Uniqueness of H). *Let $f : \mathbb{N}^+ \rightarrow \mathbb{R}$ satisfy (a) $f(mn) = f(m) + f(n)$, (b) $f(n) \geq 0$, (c) $f(n+1) - f(n) \rightarrow 0$. Then $f = c \log n$ for some $c \geq 0$.*

Proof. By the Erdős–Kátaı theorem [4, 5]. An independent proof follows from Ostrowski’s theorem: $e^{f(n)}$ is an absolute value on $\mathbb{Z}$; the condition $f \geq 0$ rules out all p -adic absolute values, leaving only the Archimedean absolute value n^c [6]. $\square$

3 Axiom I: Hilbert Space

The first axiom of quantum mechanics requires that physical states form a Hilbert space—a complete linear space equipped with an inner product. The inner product endows states with lengths and angles, making probability and interference possible. In the standard formulation, the Hilbert space is postulated a priori.

Theorem 4. *The C^* -algebraic positivity of the KMS state naturally gives rise to the inner product $\langle m|n\rangle = \delta_{mn}$ and the state space $\ell^2(\mathbb{N}^+)$.*

Proof. No inner product is assumed a priori. The construction proceeds from the BC data in four steps.

Step 1 (Positivity $\Rightarrow$ sesquilinear form). Each KMS state φ_β is a positive linear functional on the BC algebra $\mathcal{A}$: $\varphi_\beta(a^*a) \geq 0$ for all $a \in \mathcal{A}$. This positivity defines a sesquilinear form on $\mathcal{A}$:

$$\langle a, b \rangle_\varphi = \varphi_\beta(b^*a). \quad (1)$$

This form is automatically conjugate-symmetric ($\langle a, b \rangle_\varphi = \overline{\langle b, a \rangle_\varphi}$) and positive semi-definite ($\langle a, a \rangle_\varphi \geq 0$). This is a mathematical consequence of C^* -positivity.

Why $p = 2$: $\varphi(b^*a)$ is a sesquilinear form—a map that is simultaneously linear in a and antilinear in b . A sesquilinear form is precisely the definition of an inner product, and the completion of an inner product space is L^2 (a Hilbert space). The exponent $p = 2$ arises from the structure of the sesquilinear form, i.e., as a natural consequence of the positivity $\varphi(a^*a) \geq 0$ of a C^* -algebraic state.

Step 2 (Quotient and completion $\Rightarrow$ Hilbert space). Quotienting by the null space $\{a : \langle a, a \rangle_\varphi = 0\}$ and completing yields the Hilbert space $\mathcal{H}_\varphi$. For the BC system, the KMS state at $\beta > 1$ gives $\varphi_\beta(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta) > 0$ for all n , so no nonzero basis vector is a null element. The resulting Hilbert space is $\mathcal{H}_\varphi = \ell^2(\mathbb{N}^+)$.

Step 3 (Orthogonality). Under the GNS inner product $\langle a, b \rangle_\varphi = \varphi_\beta(b^*a)$, H as an element of the BC algebra satisfies symmetry: $\varphi_\beta((Hg)^*f) = \varphi_\beta(g^*Hf)$, since H is self-adjoint ($H^* = H$, $\log n \in \mathbb{R}$). In the GNS representation, this is equivalent to:

$$\langle Hf, g \rangle_\varphi = \sum_n (\log n) \overline{f(n)} g(n) = \langle f, Hg \rangle_\varphi. \quad (2)$$

Since H has simple spectrum (for $m \neq n$, $\log m \neq \log n$),

$$(\log m - \log n) \langle m|n \rangle_\varphi = 0 \implies \langle m|n \rangle_\varphi = 0 \quad (m \neq n). \quad (3)$$

Step 4 (Normalization $w_n = 1$). The GNS construction establishes the existence of the inner product and orthogonality (Steps 1–3), but the norm of each basis vector has not yet been fixed. The GNS inner product gives $\langle n|n \rangle_\varphi = \varphi_\beta(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta)$, which is not the standard normalization. Standard normalization is achieved by rescaling: define $|\tilde{n}\rangle = \sqrt{\zeta(\beta)/n^{-\beta}} |n\rangle$, so that $\langle \tilde{m}|\tilde{n} \rangle_\varphi = \delta_{mn}$.

This rescaling is determined by the partition function $Z = \zeta$. In the standard basis $\{|\tilde{n}\rangle\}$, the partition function is $Z(\beta) = \sum_n \langle \tilde{n}|e^{-\beta H}|\tilde{n} \rangle_\varphi = \sum_n n^{-\beta} = \zeta(\beta)$, with coefficients exactly equal to 1. This is a direct computation: $\langle \tilde{n}|e^{-\beta H}|\tilde{n} \rangle = n^{-\beta}$, with coefficient exactly 1.

In summary: the GNS construction yields the structure (orthogonality), and $Z = \zeta$ yields the normalization ($w_n = 1$). These two aspects are compatible but distinct.

Conclusion. $\langle m|n \rangle = \delta_{mn}$. The state space is $\ell^2(\mathbb{N}^+)$ —constructed, not postulated. $\square$

In the BC framework, the Hilbert space $\ell^2(\mathbb{N}^+)$ arises naturally from the C^* -positivity condition $\varphi(a^*a) \geq 0$, and the discrete basis vectors yield a complete state space upon completion.

4 Axiom II: Self-Adjoint Observables

The second axiom of quantum mechanics requires that observables—energy, position, momentum, and other physical quantities—be represented by self-adjoint operators. Self-adjointness ensures that measurement outcomes are real numbers and that eigenstates are orthogonal. In the standard formulation, this is an additional assumption independent of the state space.

Theorem 5. *The observables of the BC system—operators commuting with H and generated by arithmetic functions—are all self-adjoint operators.*

Proof. Self-adjointness of H . The domain is $\text{dom}(H) = \{f \in \ell^2 : \sum_n (\log n)^2 |f(n)|^2 < \infty\}$. Symmetry: for $f, g \in \text{dom}(H)$, $\langle Hf, g \rangle = \langle f, Hg \rangle$. The set $\{|n\rangle\}$ constitutes a complete eigenbasis with real eigenvalues $\log n$; a symmetric operator with a complete real eigenbasis is self-adjoint (deficiency indices $(0, 0)$) [7].

Simple spectrum. $\log m = \log n \Leftrightarrow m = n$ ($\log$ is injective on $\mathbb{N}^+$), so each eigenspace of H is one-dimensional.

Commutant. If a bounded operator A commutes with H , then the simplicity of the spectrum forces A to preserve each one-dimensional eigenspace, so $A|n\rangle = a_n|n\rangle$ (i.e., A must be diagonal).

Arithmetic observables are self-adjoint. The diagonal operators naturally arising from the BC system include: $H(n) = \log n$ (energy, $\log n \in \mathbb{R}_{\geq 0}$), $\hat{n}_p(n) = v_p(n)$ (p -channel occupation number, $v_p \in \mathbb{Z}_{\geq 0}$), $\hat{N}(n) = \Omega(n) = \sum_p v_p(n)$ (total number of prime factors, $\Omega \in \mathbb{Z}_{\geq 0}$), and $\mu_m \mu_m^*$ (projections with eigenvalues in $\{0, 1\}$). All of these operators have non-negative real eigenvalues—not by assumption, but as a direct consequence of the ranges of the arithmetic functions $\log$, v_p , and Ω . Diagonal operators with real eigenvalues are self-adjoint. $\square$

Quantum mechanics requires that measurement outcomes of observables be real numbers. In the BC framework this requires no additional assumption. Arithmetic functions on the integers— $\log n$, $v_p(n)$, $\Omega(n)$ —take real values naturally. Self-adjointness is not a constraint imposed on operators, but a property inherent to arithmetic itself: numbers obtained by counting are naturally real.

Remark 6 (Position and momentum). *The observable algebra of the BC system is generated by functions of $H = \log n$, all of which commute with H . The non-commuting position and momentum operators of standard quantum mechanics ($[\hat{x}, \hat{p}] = i\hbar$) require additional structure from a continuous space and lie outside the scope of this paper.*

5 Axiom III: Born Rule

The third axiom of quantum mechanics—the Born rule—relates the components of a state vector to measurement probabilities: $P(n|\psi) = |\langle n|\psi \rangle|^2$. In the standard formulation, this requires the introduction of measurement, observers, and wavefunction collapse.

Theorem 7. *The overlap between distinct KMS states of the BC algebra uniquely determines the Born rule $P(n|\psi) = |\langle n|\psi \rangle|^2$.*

Proof. This proof derives the Born rule from the relational structure of KMS states—without invoking measurement, observers, or collapse.

Step 1 (KMS states as relational objects). For $\beta > 1$, the BC system possesses a family of KMS states $\{\varphi_{\beta, \alpha}\}_{\alpha \in \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})}$, related by Galois action [1]. All states agree on diagonal quantities:

$$\varphi_{\beta, \alpha}(|n\rangle\langle n|) = n^{-\beta} / \zeta(\beta) \quad (\text{independent of } \alpha). \quad (4)$$

They differ on off-diagonal quantities (phases): $\varphi_{\beta, \alpha}(\zeta_n) = \alpha(\zeta_n)$.

Step 2 (Orthogonality $\rightarrow$ functional form of probability). Let $|\psi\rangle = \sum_n c_n |n\rangle$ be a state in $\ell^2(\mathbb{N}^+)$. By orthogonality (Theorem 4, Step 3),

$$\langle n|\psi \rangle = \sum_m c_m \langle n|m \rangle = \sum_m c_m \delta_{nm} = c_n. \quad (5)$$

Orthogonality eliminates all terms with $m \neq n$. Hence, the *entire information* of $|\psi\rangle$ in the direction $|n\rangle$ is c_n —no other coefficient $\{c_m\}_{m \neq n}$ can influence this projection.

Now let $P(n|\psi)$ be the probability for $|\psi\rangle$ in the direction $|n\rangle$. By the above argument, $P(n|\psi)$ can only depend on c_n . By Step 1, different KMS states agree on diagonal quantities but differ on phases. Physical observables—including probabilities—should not depend on the choice of Galois element α , since α labels equivalent descriptions of the same physical system. The diagonal quantity $\varphi_{\beta,\alpha}(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta)$ is independent of α ; while the phase of c_n varies with α ($\varphi_{\beta,\alpha}(\zeta_n) = \alpha(\zeta_n)$). Therefore, probability, as an invariant shared by all KMS states, can only depend on $|c_n|$:

$$P(n|\psi) = f(|c_n|). \quad (6)$$

Since H has simple spectrum (log is injective on $\mathbb{N}^+$), $\{|n\rangle\}$ is the unique eigenbasis. Non-contextuality holds automatically here: simple spectrum $\rightarrow$ unique eigenbasis $\{|n\rangle\} \rightarrow$ no freedom to choose a different basis $\rightarrow$ non-contextuality.

That f is independent of n follows from normalization. First, the two-component case: for any m, n, n' , take $|\psi\rangle = \alpha|m\rangle + \beta|n\rangle$ and $|\psi'\rangle = \alpha|m\rangle + \beta|n'\rangle$; normalization gives $f(|\alpha|) + f_n(|\beta|) = 1 = f(|\alpha|) + f_{n'}(|\beta|)$, so $f_n = f_{n'}$. The general case follows by induction: assume f is independent of n for states with k components. Take a $(k+1)$ -component state $|\psi\rangle = \alpha|m\rangle + \beta|\psi_{\text{rest}}\rangle$, where $|\psi_{\text{rest}}\rangle$ has k components. Normalization gives $f(|\alpha|) + \sum_j f(|c_{n_j}|) = 1$; by the induction hypothesis, $\sum_j f(|c_{n_j}|)$ is independent of the specific choice of $\{n_j\}$, so $f(|\alpha|)$ is also independent of m . The induction is complete.

Step 3 (C^* -positivity constrains f). The function $f : [0, 1] \rightarrow [0, 1]$ satisfies the constraints listed below.

Step 4 (C^* -positivity constrains f). The function $f : [0, 1] \rightarrow [0, 1]$ satisfies:

- (a) $f(x) \geq 0$ (C^* -state positivity: $\varphi(a^*a) \geq 0$);
- (b) $\sum_n f(|c_n|) = 1$ (normalization: $\varphi_\beta(I) = 1$);
- (c) $f(0) = 0$ (orthogonality: $\langle m|n\rangle = 0$ for $m \neq n$, Theorem 4);
- (d) $\sum_{n \in A} f(|c_n|) = f_A$ holds for mutually exclusive outcomes ($\mathbb{N}^+$ is discrete; on a discrete space, $P(A) = \sum_{n \in A} P(n)$ is the definition of event probability).

Step 5 (Determining f on a dense set—using only the eigenbasis). Equal-amplitude states $|\psi\rangle = \frac{1}{\sqrt{k}} \sum_{j=1}^k |n_j\rangle$ give $f(1/\sqrt{k}) = 1/k$ by (b).

Now take j basis states with amplitude $x_0 = 1/\sqrt{k}$ and one remainder state:

$$|\psi\rangle = \frac{1}{\sqrt{k}} \sum_{i=1}^j |n_i\rangle + \sqrt{1-j/k} |n_{j+1}\rangle, \quad (7)$$

which is normalized: $j/k + (1-j/k) = 1$. By (b):

$$j \cdot f(1/\sqrt{k}) + f(\sqrt{1-j/k}) = 1 \implies f(\sqrt{m/k}) = m/k, \quad (8)$$

where $m = k - j$ ranges over $0, 1, \dots, k$. The set $\{\sqrt{m/k} : 0 \leq m \leq k, k \in \mathbb{N}^+\}$ is dense in $[0, 1]$. On this dense set, $f(x) = x^2$.

No coarse-graining, no basis change, no subspace projection. The entire argument uses only eigenstates $\{|n\rangle\}$ and the normalization condition (b).

Step 6 (Cauchy equation + non-negativity $\Rightarrow f(a) = a^2$ for all a). Define $h(t) = f(\sqrt{t})$ ($t \in [0, 1]$), so that $f(x) = h(x^2)$.

Cauchy equation. Take three distinct eigenstates $|1\rangle, |2\rangle, |3\rangle$ and construct two states:

$$|\psi_1\rangle = \sqrt{s} |1\rangle + \sqrt{t} |2\rangle + \sqrt{1-s-t} |3\rangle, \quad (9)$$

$$|\psi_2\rangle = \sqrt{s+t} |1\rangle + \sqrt{1-s-t} |2\rangle, \quad (10)$$

where $s, t \geq 0$ and $s + t \leq 1$. Both are normalized finite-component states. By condition (b):

$$h(s) + h(t) + h(1-s-t) = 1, \quad (11)$$

$$h(s+t) + h(1-s-t) = 1. \quad (12)$$

Subtracting:

$$h(s+t) = h(s) + h(t) \quad (s, t \geq 0, s+t \leq 1). \quad (13)$$

This is the Cauchy functional equation on $[0, 1]$, derived entirely using finite-component states.

Solution on $\mathbb{Q}$. By Step 5, $f(\sqrt{m/k}) = m/k$, i.e., $h(m/k) = m/k$ for all rationals $m/k \in [0, 1]$.

Extension to all of $[0, 1]$ via non-negativity. For any $\alpha \in [0, 1]$ (possibly irrational):

Lower bound. For any rational r with $0 \leq r < \alpha$: $h(\alpha) = h(r) + h(\alpha-r) = r + h(\alpha-r) \geq r$, since $h(\alpha-r) \geq 0$ (condition (a)). Taking the supremum over all such r : $h(\alpha) \geq \alpha$.

Upper bound. For any rational r with $\alpha < r \leq 1$: $h(r) = h(\alpha) + h(r-\alpha)$, i.e., $r = h(\alpha) + h(r-\alpha)$, so $h(\alpha) = r - h(r-\alpha) \leq r$, since $h(r-\alpha) \geq 0$ (condition (a)). Taking the infimum over all such r : $h(\alpha) \leq \alpha$.

Combining: $h(\alpha) = \alpha$ for all $\alpha \in [0, 1]$.

Conclusion. $f(x) = h(x^2) = x^2$ for all $x \in [0, 1]$. Therefore $P(n|\psi) = |c_n|^2 = |\langle n|\psi \rangle|^2$.

Inputs used in this step: the Cauchy equation uses only normalization (b) for finite-component states; the extension uses only non-negativity (a) (C^ -positivity) and the density of $\mathbb{Q}$ in $\mathbb{R}$. No monotonicity, continuity, or infinite-component states are required.* $\square$

Remark 8 (Sources of all conditions in the BC system). • **(a) Positivity:** $\varphi(a^*a) \geq 0$ is the defining property of a C^* -state, uniquely determined by the BC data (Remark 2).

- **(b) Normalization:** $\varphi_\beta(I) = \zeta(\beta)/\zeta(\beta) = 1$.
- **(c) Orthogonality:** Theorem 4, Step 3.
- **(d) Additivity:** The inner product is the unique GNS inner product; it defines a metric on the state space; the overlap functional in a metric space is constructively additive.
- **Non-contextuality:** Orthogonality $\langle m|n \rangle = 0$ eliminates all $m \neq n$ terms in $\langle n|\psi \rangle = c_n$ (Step 3).
- **Independence of f from n :** normalization (Step 3).
- **No basis-independence needed:** Step 5 uses only eigenstates $\{|n\rangle\}$ and normalization—no coarse-graining.

The Born rule $P = |c_n|^2$ [16] in quantum mechanics is usually regarded as an independent postulate—it tells us the probability of obtaining a given outcome upon measurement. The derivation above shows that probability is not a statement about “what happens upon measurement.” It is the overlap between two states—how much of $|\psi\rangle$ is visible from the perspective of the state $|n\rangle$. This overlap is uniquely determined by C^* -positivity to be $|c_n|^2$ —no more and no less. Probability is an objective relation between two states—not the output of a measurement, not the belief of an observer, not an intrinsic property of a particle.

6 Axiom IV: Unitary Time Evolution

The fourth axiom of quantum mechanics requires that the time evolution of an isolated system be described by a unitary group $U(t) = e^{-iHt}$ —preserving probability and inner products. In the standard formulation, the Hamiltonian H is an additional input. Connes–Rovelli [11] proposed that time may arise from the modular automorphism group.

Sign convention. The algebraic automorphism $\sigma_t(\mu_m) = m^{it}\mu_m$ (Definition 1(v)) corresponds to the Heisenberg picture; the Schrödinger-picture unitary is $U(t) = e^{-itH}$, giving $U(t)|n\rangle = n^{-it}|n\rangle = e^{-it \log n}|n\rangle$. The two are related by $\sigma_t(A) = U(-t)AU(t)$.

Theorem 9 (Without additional assumptions). *The BC system possesses a unique strongly continuous one-parameter unitary group*

$$U(t)|n\rangle = e^{-it \log n}|n\rangle, \quad t \in \mathbb{R}, \quad (14)$$

with generator $H = \log n$: $U(t) = e^{-itH}$. The one-parameter automorphism group σ_t and the time parameter t are both given by the arithmetic data of the BC system.

Proof. Unitarity. $\|U(t)f\|^2 = \sum_n |e^{-it \log n}|^2 |f(n)|^2 = \|f\|^2$. $U(t)^* = U(-t)$, $U(t)^*U(t) = I$.

Group law. $U(s)U(t)|n\rangle = e^{-i(s+t) \log n}|n\rangle = U(s+t)|n\rangle$. $U(0) = I$.

Strong continuity. The action of $U(t)$ on each state $|n\rangle$, namely $U(t)|n\rangle = n^{it}|n\rangle$, is a continuous function of t (at the discrete level). Upon completion to $\ell^2(\mathbb{N}^+)$, pointwise continuity upgrades to ℓ^2 strong continuity: $\|U(t)f - f\|^2 = \sum_n |e^{-it \log n} - 1|^2 |f(n)|^2$; since $|e^{i\theta} - 1| \leq 2$, the integrand is dominated by $4|f(n)|^2 \in \ell^1$, and the dominated convergence theorem [7] gives $\|U(t)f - f\| \rightarrow 0$. Strong continuity is not an additional assumption—it is an automatic consequence of pointwise continuity at the discrete level after completion.

Generator. $-i \frac{d}{dt} U(t)|n\rangle|_{t=0} = (\log n)|n\rangle = H|n\rangle$. Therefore $U(t) = e^{-itH}$ (H is self-adjoint by Theorem 5).

Uniqueness. Suppose σ'_t is another continuous one-parameter $*$ -automorphism group of the BC algebra. We prove $\sigma'_t = \sigma_t$ in five steps.

Step 1: σ'_t preserves each generator. The generators μ_m ($m \in \mathbb{N}^+$) of the BC algebra are labeled by a discrete set. Since σ'_t is a strongly continuous path starting at the identity ($\sigma'_0 = \text{id}$), a continuous path from the identity cannot permute discrete objects; therefore σ'_t must preserve each μ_m up to a scalar:

$$\sigma'_t(\mu_m) = \lambda_m(t) \mu_m, \quad \lambda_m(t) \in \mathbb{C}.$$

Step 2: $|\lambda_m(t)| = 1$. μ_m is an isometry ($\mu_m^* \mu_m = I$). A $*$ -automorphism preserves the norm, so $|\lambda_m(t)| = 1$ for all m, t .

Step 3: $\lambda_m(t)$ is completely multiplicative. The multiplicative relation $\mu_m \mu_n = \mu_{mn}$ is a defining relation of the BC algebra, and σ'_t as a $*$ -automorphism must preserve it:

$$\lambda_{mn}(t) = \lambda_m(t) \lambda_n(t).$$

The group law $\sigma'_{t+s} = \sigma'_t \circ \sigma'_s$ gives:

$$\lambda_m(t+s) = \lambda_m(t) \lambda_m(s).$$

Step 4: $\lambda_m(t) = m^{ict}$. Complete multiplicativity ($\lambda_{mn} = \lambda_m \lambda_n$) together with unitarity ($|\lambda_m| = 1$) implies $\lambda_m(t) = m^{ig(t)}$ for some real-valued function $g(t)$. The group law $\lambda_m(t+s) = \lambda_m(t) \lambda_m(s)$ gives the Cauchy functional equation $g(t+s) = g(t) + g(s)$. Continuity yields the unique solution $g(t) = ct$ for some $c \in \mathbb{R}$. Therefore $\lambda_m(t) = m^{ict}$, corresponding to the Hamiltonian $H' = cH = c \log n$.

Step 5: $c = 1$. The partition function $Z'(\beta) = \sum_n n^{-c\beta} = \zeta(c\beta)$. The BC system requires $Z(\beta) = \zeta(\beta)$, so $\zeta(c\beta) = \zeta(\beta)$ for all $\beta > 1$. Since ζ is strictly decreasing on $(1, \infty)$ (each term $n^{-\beta}$ is strictly decreasing), we must have $c\beta = \beta$, i.e., $c = 1$. Hence $\sigma'_t = \sigma_t$. $\square$

The derivation above shows that time is not an externally introduced physical concept. The BC system intrinsically possesses a unique one-parameter dynamics: $\sigma_t(f)(n) = n^{it} f(n)$. The existence of this dynamics follows from the real-valuedness of $H = \log n$, and its uniqueness from the strict monotonicity of $\zeta(\beta)$. Stone's theorem and Tomita–Takesaki theory are not needed—the BC system naturally encodes the existence and uniqueness of time, consistent with the KMS framework of Haag–Hugenholtz–Winnink [12]. Bratteli–Robinson [17] provides a systematic treatment of operator algebras and quantum statistical mechanics.

7 Axiom V: Tensor Product

The fifth axiom of quantum mechanics requires that the state space of a composite system be the tensor product of the state spaces of the constituent subsystems. In the standard formulation, the tensor product structure is an independent assumption.

Theorem 10 (Without additional assumptions). *The fundamental theorem of arithmetic gives a unitary isomorphism*

$$\ell^2(\mathbb{N}^+) \cong \bigotimes_{p \text{ prime}} \ell^2(\mathbb{Z}_{\geq 0}), \quad (15)$$

where the right-hand side is the von Neumann infinite tensor product [3] stabilized by $|0\rangle_p$.

Proof. Correspondence of bases. Unique factorization $n = \prod_p p^{a_p}$ gives a bijection $\mathbb{N}^+ \leftrightarrow \bigoplus_p \mathbb{Z}_{\geq 0}$, inducing $|n\rangle \leftrightarrow \bigotimes_p |a_p(n)\rangle_p$. Only finitely many $a_p \neq 0$ ($|0\rangle_p$ serves as the stabilizing vector for almost all p).

Orthogonality. $\langle m|n\rangle = \delta_{mn} \Leftrightarrow \prod_p \delta_{a_p(m), a_p(n)} = \delta_{mn}$.

Completeness. $\{|n\rangle\}$ and $\{\bigotimes_p |a_p\rangle_p\}$ are complete orthonormal bases for their respective spaces. A bijection between complete orthonormal bases extends uniquely to a unitary isomorphism. $\square$

Remark 11. *The Hamiltonian decomposes as $H = \sum_p (\log p) \hat{n}_p$ —the sum of energies of independent Bose channels. The Euler product $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$ is the manifestation of this tensor product decomposition in the partition function. The correspondence between positive integers and Fock states of prime channels was independently observed by Spector [8] and Julia [9] in the framework of arithmetic quantum field theory. The contribution of the present paper is to place this observation within a complete axiomatic chain derived from the BC system. Connes–Marcolli [13] provides a broader treatment of noncommutative geometry and its connections to number theory.*

Remark 12. *The stabilizing vector $|0\rangle_p$ corresponds to $n = 1$ (all $a_p = 0$), i.e., the Fock vacuum of the BC system. In rings where unique factorization fails (e.g., $\mathbb{Z}[\sqrt{-5}]$, where $6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$), the bijection of bases does not hold, and the tensor product decomposition no longer exists [3]. We note that Theorem 10 establishes a subsystem decomposition of the BC state space, rather than the general quantum mechanical principle of composing arbitrary systems via tensor products; the latter requires additional physical input.*

The derivation above shows that the tensor product structure arises directly from the fundamental theorem of arithmetic—each prime p corresponds to an independent channel, and the independence of subsystems is precisely the independence of prime channels. It is worth noting that this structure naturally decomposes the state space into tensor product factors indexed by distinct primes, and such a decomposition could in principle provide a natural description of internal symmetries in particle physics.

8 Discussion

8.1 Summary

Axiom	Physical requirement	BC correspondence	Thm.
I	Hilbert space	C^* -positivity $\rightarrow$ GNS construction $\rightarrow \ell^2(\mathbb{N}^+)$	4
II	Self-adjoint observables	Self-adjoint part of C^* -algebra + real range of arithmetic functions	5
III	Born rule	KMS overlap + orthogonality + normalization $\rightarrow c_n ^2$	7
IV	Unitary evolution	Strict monotonicity of $\zeta \rightarrow c = 1 \rightarrow$ unique σ_t	9
V	Tensor product	Unique factorization $\rightarrow$ prime channel decomposition	10

The table above summarizes the corresponding structure of each axiom within the BC system. The BC system is capable of encoding the core content of all five axioms.

8.2 Open Problems

The state space $\ell^2(\mathbb{N}^+)$ of the BC system is discrete: basis states are labeled by positive integers, the energy spectrum $\{\log n\}$ is a discrete set, and the minimum energy gap is $\log 2$. This is structurally distinct from the continuous Hilbert space $L^2(\mathbb{R}^d)$ widely used in contemporary physics.

However, the discrete structure does not contradict the observational results of continuous physics. The energy spacing between adjacent states, $\kappa(n) = \log(1+1/n)$, decreases monotonically to zero with n — when n is sufficiently large, the discrete spacing is far below any experimentally resolvable scale, and the macroscopic behavior is indistinguishable from that of a continuous space. The completion process (§3) has already demonstrated how discrete basis vectors naturally give rise to a complete Hilbert space; the strong continuity (§6) further shows that pointwise continuity at the discrete level automatically upgrades to ℓ^2 continuity upon completion.

On the other hand, the underlying discreteness may have positive significance. Several fundamental problems that afflict physics in the continuous framework— ultraviolet divergences, the black hole information paradox, spacetime singularities— are all related to the existence of infinitesimally small distances. Discrete structures offer a natural possibility for resolving these problems.

The results of this paper show that the BC system can precisely encode all five axioms of quantum mechanics. The precise relationship between the discrete and the continuous is a direction worthy of further investigation.

Data availability

No datasets were generated or analyzed during the current study. All results are obtained by analytic derivation.

Conflict of interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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